UNIT – 4

ASSESSMENT – 3

SOLUTIONS

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EXPERIMENT 1:

The **Decision Tree algorithm** is a supervised machine-learning algorithm used for classification.

For this experiment, the **Iris dataset** is used.

Features

The Iris dataset contains:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target

The target is the Iris flower species:

0 → Setosa

1 → Versicolor

2 → Virginica

Splitting Criterion

The **Gini Index** is used to determine the best feature for splitting.

The feature that gives the best reduction in impurity becomes the root node.

Train-Test Split

80% → Training

20% → Testing

EXPERIMENT 2:

We use a **4 × 4 Grid World**.

The states are numbered:

```
 0   1   2   3
 4   5   6   7
 8   9  10  11
12  13  14  15
```

Let's define:

Start = 0

Goal = 15

Obstacle = 6

Actions

0 → Up

1 → Down

2 → Left

3 → Right

Rewards

Reach Goal → +10

Normal step → -1

Obstacle → -5

Hyperparameters

α (Learning Rate) = 0.1

γ (Discount Factor) = 0.9

ε (Exploration Rate) = 0.2

The Q-Learning equation is: